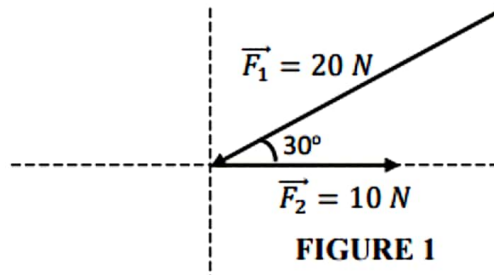


1. a) Calculate the momentum, p of object in SI unit if mass, $m = 2$ kg and velocity, $v = 300 \text{ km h}^{-1}$
- b) Two forces $F_1 = 20 \text{ N}$ and $F_2 = 10 \text{ N}$ are acting on Block A. **Figure 1** below. Determine the magnitude of the resultant force acting on Block A.



NO	ANSWER SCHEME
1 (a)	$v = \left(\frac{300\cancel{\text{km}}}{1\cancel{\text{h}}}\right)\left(\frac{1000\cancel{\text{m}}}{1\cancel{\text{km}}}\right)\left(\frac{1\cancel{\text{h}}}{3600\text{s}}\right)$ $v = 83.33\text{ms}^{-1}$ $p = mv$ $p = (2)(83.33) = 166.67 \text{ kgms}^{-1}$
(b)	$F_{1x} = -20 \cos 30 = -17.32 \text{ N}$ $F_{1y} = -20 \sin 30 = -10 \text{ N}$ $\sum F_x = F_{1x} + F_{2x} = -17.32 + 10 = -7.32 \text{ N}$ $\sum F_y = F_{1y} = -20 \sin 30 = -10 \text{ N}$ $F_{\text{nett}} = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = \sqrt{(-7.32)^2 + (-10)^2}$ $= 12.39 \text{ N}$ $\theta = \tan^{-1} \frac{\sum F_y}{\sum F_x} = \tan^{-1} \frac{-10}{-7.32}$ $= 53.80^\circ, \text{ below negative } x - \text{axis}$
	TOTAL

- (i) final velocity of the lorry after 25 s.
- (ii) acceleration of the lorry in 25 s.
- (iii) If the lorry then slowing down its speed at rate 0.60 m travel before it stop.

(b) A ball is thrown vertically upward with speed of 19.0 m s⁻¹ from building. Calculate the

- (i) maximum height reached by the ball from the released
- (ii) time taken by the ball to reach the ground

NO	ANSWER SCHEME
2(a)	(i) $s = \frac{1}{2}(u + v)t$ $300 = \frac{1}{2}(0 + v)(25)$ $v = 24 \text{ ms}^{-1}$ 3 / 11 ⋮
	(ii) $v = u + at$ $24 = 0 + a(25)$ $a = 0.96 \text{ ms}^{-2}$
	(iii) $a = -0.60 \text{ ms}^{-2}$ $v^2 = u^2 + 2as$ $(0)^2 = (24)^2 + 2(-0.60)s$ $s = 480 \text{ m}$
2(b)	(i) $v = 0 \text{ ms}^{-1}$ $v^2 = u^2 - 2gs$ $(0)^2 = (19.0)^2 - 2(9.81)s$ $s = 18.40 \text{ m}$
	(ii) $s = ut - \frac{1}{2}gt^2$ $-10 = (19.0)t - \frac{1}{2}(9.81)t^2$ $t_1 = -0.47 \text{ s(neglected)} \quad \text{or} \quad t_2 = 4.34 \text{ s}$

	$t = 4.34 \text{ s}$
	TOTAL

3. (a) A 8 kg moves towards the west with a momentum of 30 kgms⁻¹ to the east acts on the body for a period of 15 s. determine the

- (i) the impulse of the force
- (ii) the change in the momentum of a body
- (iii) the final momentum of the body
- (iv) the final velocity of the body

(b)

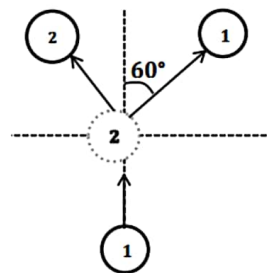


Figure 3

A ball 1 with mass 0.15kg moving to the north at speed 5 ms⁻¹ stationary ball 2 whose mass is 0.20 kg. After collide the ball 1 speed 9 ms⁻¹. Calculate the velocity of ball 2 after collision.

NO	ANSWER SCHEME
3(a)	$J = Ft = (20)(15) = 300Ns$
(i)	
(ii)	$J = \Delta P = 300Ns$
(iii)	$J = p_f - p_i$ $300 = p_f - (-30)$ $p_f = 270 \text{ kgms}^{-1}$
(iv)	$p_f = mv$ $270 = 8v$

	$v = 33.75 \text{ ms}^{-1}$
(b)	$\Sigma p_{ix} = \Sigma p_{fx}$ $m_1 u_{1x} + m_2 u_{2x} = m_1 v_{1x} \sin 60^\circ + m_2 v_{2x}$ $0 = (0.15)(9)(\sin \sin 60^\circ) + 0.20 v_{2x}$ $v_{2x} = -5.85 \text{ ms}^{-1}$ $m_1 u_{1y} + m_2 u_{2y} = m_1 v_{1y} \cos 60^\circ + m_2 v_{2y}$ $0.15)(5) + 0 = (0.15)(9)(\cos \cos 60^\circ) + 0.20 v_{2y}$ $v_{2y} = 0.38 \text{ ms}^{-1}$ $v = \sqrt{v_{2x}^2 + v_{2y}^2}$ $= \sqrt{(-5.85)^2 + (0.38)^2}$ $v_2 = 5.86 \text{ ms}^{-1}$
	TOTAL

4. (a) **Figure 4.1** show two block A and B connected to a massless string through smooth pulleys. Block A is placed on a rough inclined plane with coefficient of kinetic friction is 0.6 with a mass of 25kg moving up the surface. Block B moving downward experiences an acceleration of 5 ms^{-2} .

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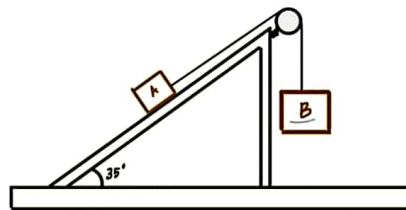


Figure 4.1

- Draw free body diagram to show all forces that act on block A.
 - Calculate the frictional force acting on the block A.
 - Calculate the tension of the rope that connecting block A and B.
 - Calculate the mass of block B.
- (b) Two boxes moving at a frictionless horizontal surface are connected by a string as shown in **Figure 4.2**, where $m_1 = 12 \text{ kg}$ and $m_2 = 25 \text{ kg}$. A force is applied to the m_2 box with an angle of 45° with the horizontal surface.

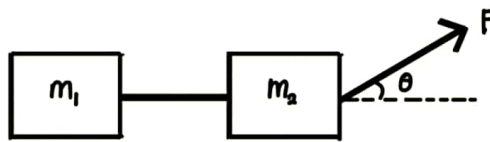


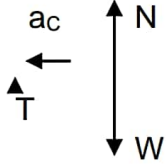
Figure 4.2

- (i) Find the acceleration of each box
- (ii) Find the tension in the string

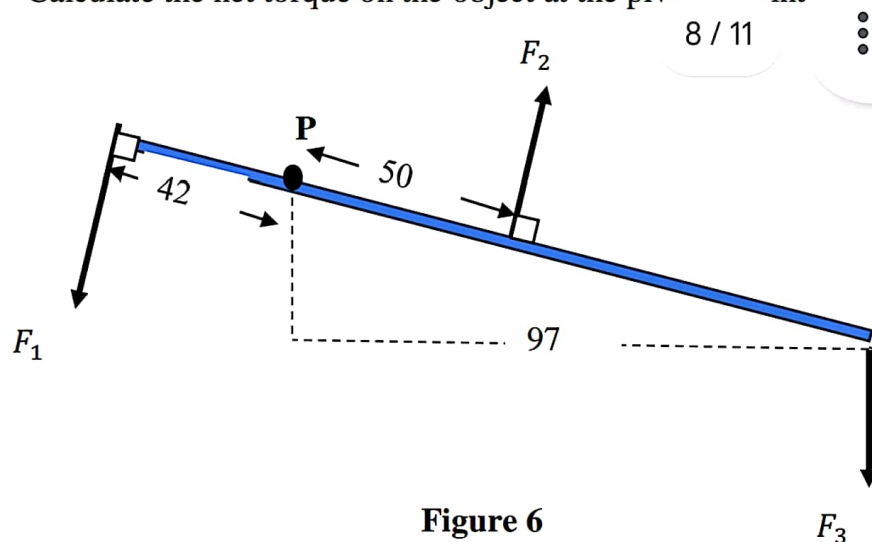
NO	ANSWER SCHEME
1 a)	
b)	$\sum F_y = 0$ $N_A - m_A g \cos \theta = 0$ $N_A = (25)(9.81) \cos 35$ $N_A = 200.90 \text{ N}$ $f_k = \mu_k N$ $= (0.6)(200.90)$ $= 120.54 \text{ N}$
c)	$\sum F_x = m_A a$ $T - f_k - m_A g \sin \theta = m_A a$ $T - (120.54) - (25)(9.81) \sin 35 = (25)(0.5)$ $T = 273.71 \text{ N}$
d)	$\sum F_y = m_B a$ $W_B - T = m_B a$ $m_B (9.81) - T = m_B (0.5)$ $m_B = 29.4 \text{ kg}$

2 a)	$\sum F_x = m_2 a$ $F \cos 45 - T = m_2 a$ $60 \cos 45 - T = 25a \quad \text{_____ (1)}$ $F_x = m_1 a$ $T = m_1 a$ $T = 12a \quad \text{_____ (2)}$ $(2) \text{ into } (1), \quad 60 \cos 45 - 12a = 25a$ $a = 1.15 \text{ ms}^{-2}$ Or $\sum F_x = ma$ $F_x = ma$ $F \cos 45 = (12 + 25)a$ $a = 1.15 \text{ ms}^{-2}$
b)	$T = 12a$ $T = 13.8 \text{ N}$
	TOTAL

- (a) An object of mass 450 g moves in a circle of $7/11$ m radius. Calculate
- the angular velocity in SI unit.
 - the linear speed of the object.
- (b) An object of mass 12 g is attached to the end of a string of length 0.5 m. The object moves through 5 rounds per second in a circular motion on a horizontal surface.
- Draw a free body diagram to show all the forces acting on the object.
 - Calculate the net force acting on the object.
 - Calculate the tension in the string.

ANSWER SCHEME	
)	$\omega = \frac{5 \text{ rev}}{2 \text{ s}} = \frac{5 \times 2\pi}{2}$ $\omega = 5\pi \text{ rad s}^{-1} = 15.71 \text{ rad s}^{-1}$
)	$v = r\omega = (1.5)(5\pi)$ $v = 7.5\pi \text{ m s}^{-1} = 23.56 \text{ m s}^{-1}$
)	
)	$\sum F_x = F_c$ $F_c = mr\omega^2$ $F_c = (0.012)(0.5)(31.416)^2$ $F_c = 5.92 \text{ N}$
i)	$T = F_c = 5.92 \text{ N}$
TOTAL	

- (a) A rotating wheel moving from rest with 20 cm of radius and an angular acceleration of 3.0 rad s^{-2} for 28 s. Calculate
- The angular velocity
 - Tangential velocity
 - The angle through which the wheel has turned
- (b) Three forces are acting on a lightweight object as shown in **FIGURE 6**. Calculate the net torque on the object at the pivoted point P.



NO	ANSWER SCHEME
6 (a) (i)	$\omega_0 = 0, \alpha = 3\text{rads}^{-2}, r = 0.20\text{m}, t = 28\text{s}$ $\omega = \omega_0 + \alpha t$ $= 0 + (3)(28)$ $= 84\text{rads}^{-1}$
(ii)	$v = r\omega = (0.20)(84) = 16.8\text{ ms}^{-1}$
(iii)	$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ $= 0 + \frac{1}{2}(3)28^2$ $= 1176\text{rad}$
6 (b)	$\tau_1 = r_1 F_1 \sin \theta_1$ $= (0.42)(20)\sin 90$ $= 8.4\text{ N m}$ $\tau_2 = r_2 F_2 \sin \theta_2$ $= (0.5)(14)\sin 90$ $= 7\text{ N m}$ $\cos 38 = \frac{0.97}{r_3}$ $r_3 = 1.23\text{ m}$ $\tau_3 = r_3 F_3 \sin \theta_3$ $= -(1.23)(85)\sin 128$ $= -82.39\text{ N m}$ $\tau_{\text{nett}} = \tau_1 + \tau_2 + \tau_3$ $= 8.4 + 7 + (-82.39) = -66.99\text{ N m}$ $= 66.99\text{ N m (clockwise)}$
	TOTAL



NO	ANSWER SCHEME	M
7(a)	$A = \pi r^2$ Assalamualaikum madam $T_1 = 100^\circ\text{C}$ $T_2 = 0^\circ\text{C}$	

	$\frac{Q}{t} = -kA \frac{\Delta T}{L}$ $\frac{Q}{t} = -(385.0)(2.01 \times 10^{-4}) \frac{(0 - 100)}{(10.0 \times 10^{-2})}$ $\frac{Q}{t} = 77.41 \text{ W}$	
7(b) (i)	$v_{rms} = \sqrt{\frac{3kT}{m}}$ $645 = \sqrt{\frac{3(1.38 \times 10^{-23})(20 + 273.15)}{m}}$ $m = 2.91 \times 10^{-26} \text{ kg}$	
(b) (ii)	$v_{rmsH} = \sqrt{\frac{3RT}{M}}$ $1320 = \sqrt{\frac{3(8.31)T}{0.002}}$ $T = 139.78 \text{ K}$ $v_{rmsN} = \sqrt{\frac{3RT}{M}}$ $v_{rmsN} = \sqrt{\frac{3(8.31)(139.78)}{0.028}}$ $v_{rmsN} = 352.78 \text{ ms}^{-1}$	
7 (c) (i)	$W_{AB} = 0$ $W_{BC} = +120 \text{ J}$ $Q_{AB} = +180 \text{ J} \quad Q_{BC} = +250 \text{ J}$ $Q_{ABC} = \Delta U_{ABC} + W_{ABC}$ $Q_{AB} + Q_{BC} = \Delta U_{ABC} + W_{AB} + W_{BC}$ $180 + 250 = \Delta U + 120$ $\Delta U = 310 \text{ J}$	
(c) (ii)	Isothermal process $\Delta U = 0$ $Q = \Delta U + W$ $W = Q = -225 \text{ J}$ W negative (work done on the system)	